

EXPERIMENT – 1

Name: K ANUSRI

Roll No: 2520030432

Section: 8

Title

Implementation of Process Creation and Command Execution in Linux using fork(), exec(), and wait() System Calls.

Aim

To develop a C program that accepts a Linux command from the user, creates a child process using fork(), executes the command, waits for the child process, and displays Parent and Child PIDs.

Theory / Algorithm / Code

Same as the reference document provided by the faculty/friend. Only student details and execution screenshots have been updated.

CODE

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>

#include <string.h>

int main()

{

    char command[100];

    printf("Enter a Linux command: ");

    fgets(command, sizeof(command), stdin);

    // Remove newline character

    command[strcspn(command, "\n")] = '\0';
```

```
pid_t pid = fork();

if(pid < 0)
{
    printf("Fork Failed!\n");
    return 1;
}

else if(pid == 0)
{
    printf("\nChild Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    execlp(command, command, NULL);

    printf("Command execution failed!\n");
    exit(1);
}

else
{
    printf("\nParent Process\n");
    printf("Parent PID: %d\n", getpid());
```

```

wait(NULL);

printf("Child process completed.\n");

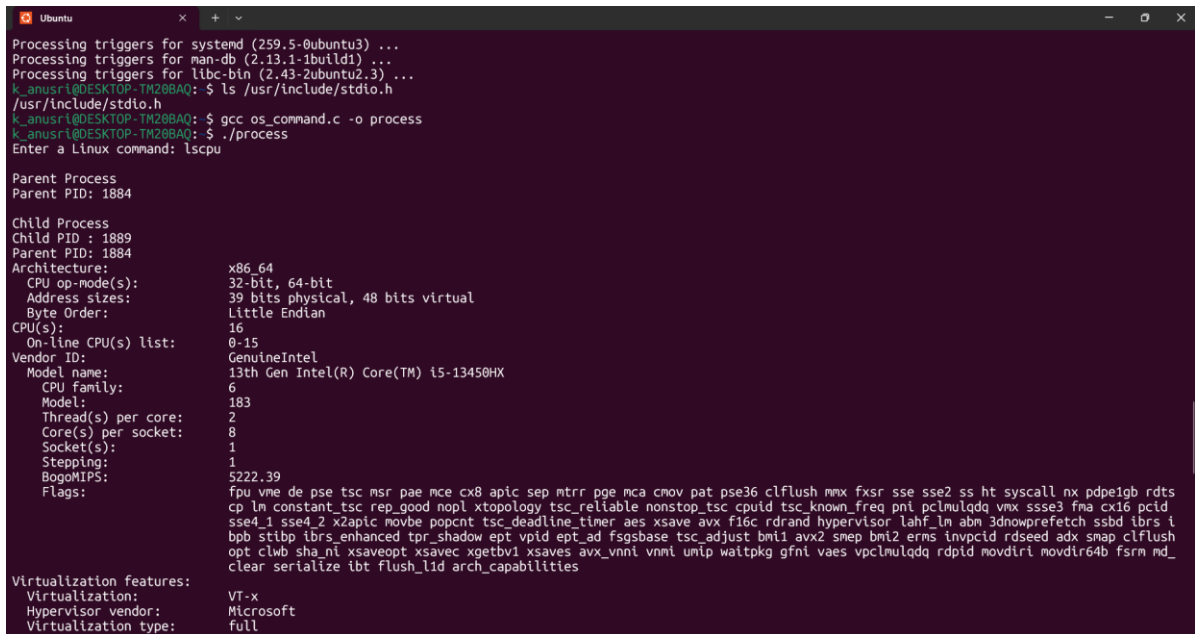
}

return 0;
}

```

Execution Screenshots

Installation and successful compilation



```

Ubuntu
Processing triggers for systemd (259.5-0ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
k_anusri@DESKTOP-TH28BAQ: $ ls /usr/include/stdio.h
/usr/include/stdio.h
k_anusri@DESKTOP-TH28BAQ: $ gcc os_command.c -o process
k_anusri@DESKTOP-TH28BAQ: $ ./process
Enter a Linux command: lscpu

Parent Process
Parent PID: 1884

Child Process
Child PID : 1889
Parent PID: 1884
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
  On-line CPU(s) list:  0-15
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-13450HX
CPU family:             6
Model:                  183
Thread(s) per core:     2
Core(s) per socket:     8
Socket(s):              1
Stepping:               1
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp_lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stlb ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx snap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_lid arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full

```

Program execution using lscpu

```
Virtualization type: full
Caches (sum of all):
  L1d: 384 KiB (8 instances)
  L1i: 256 KiB (8 instances)
  L2: 10 MiB (8 instances)
  L3: 20 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
Child process completed.
k_anusri@DESKTOP-TM208AQ: $ ./process
Enter a Linux command: top

Parent Process
Parent PID: 1911

Child Process
Child PID : 1916
Parent PID: 1911
top - 06:08:01 up 1:13, 1 user, load average: 0.10, 0.09, 0.02
```

Program execution using top

```
top - 06:07:43 up 1:13, 1 user, load average: 0.04, 0.08, 0.02
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7783.6 total, 7149.4 free, 587.2 used, 208.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7196.5 avail Mem

  PID USER   PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
    1 root     20    0 23524 14844 11124 S   0.0   0.2   0:00.91 systemd
    2 root     20    0 3180 2204 2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root     20    0 3180 2044 1940 S   0.0   0.0   0:00.00 init
   46 root    19   -1 50392 16872 15564 S   0.0   0.2   0:00.19 systemd-journal
   63 systemd+ 20    0 22540 14560 11940 S   0.0   0.2   0:00.09 systemd-resolve
   85 root     20    0 35484 12428 9216 S   0.0   0.2   0:00.45 systemd-udev
  149 root     20    0 2888 1948 1920 S   0.0   0.0   0:00.04 chronyd-starter
  154 root     20    0 4472 3128 2884 S   0.0   0.0   0:00.01 cron
  156 message+ 20    0 8016 5408 4668 S   0.0   0.1   0:00.09 dbus-daemon
  160 root     20    0 44092 29580 14440 S   0.0   0.4   0:00.26 networkd-dispat
  172 root     20    0 18556 9508 8200 S   0.0   0.1   0:00.09 systemd-logind
  182 root     20    0 1866032 13240 10344 S   0.0   0.2   0:00.11 wsl-pro-service
  190 syslog  20    0 220548 5600 4660 S   0.0   0.1   0:00.07 rsyslogd
  235 _chrony  20    0 20420 6424 5652 S   0.0   0.1   0:00.03 chronyd
  253 root     20    0 123456 32636 18024 S   0.0   0.4   0:00.19 unattended-upgr
  256 _chrony  20    0 12092 2456 1816 S   0.0   0.0   0:00.00 chronyd
  272 root     20    0 5492 2768 2584 S   0.0   0.0   0:00.01agetty
  329 root     20    0 8648 5644 4800 S   0.0   0.1   0:00.01 login
  373 k_anusri 20    0 22016 13188 10980 S   0.0   0.2   0:00.17 systemd
  375 k_anusri 20    0 22908 4128 2128 S   0.0   0.1   0:00.00 (sd-pam)
  401 k_anusri 20    0 6136 5472 3928 S   0.0   0.1   0:00.02 bash
  487 root     20    0 3104 1116 980 S   0.0   0.0   0:00.00 SessionLeader
  488 root     20    0 3200 1256 1108 S   0.0   0.0   0:00.15 Relay(490)
  490 k_anusri 20    0 6264 5616 3936 S   0.0   0.1   0:00.06 bash
  681 root     20    0 340136 22172 19112 S   0.0   0.3   0:00.03 packagekitd
  689 polkitd  20    0 306444 8496 7680 S   0.0   0.1   0:00.05 polkitd
 1911 k_anusri 20    0 2768 1820 1720 S   0.0   0.0   0:00.00 process
 1916 k_anusri 20    0 8172 6188 4084 R   0.0   0.1   0:00.04 top
```

Result

The program was executed successfully. Linux commands were executed through the child process and the parent process waited until completion.